

NAME: _____

- Exam is open-book and open-note. You are allowed to use internet, however, you are NOT allowed to talk to, text or email anybody during the exam.
- You need to submit your excel spreadsheet on BB before 11.15am. Make sure you save your file frequently and save as xlsx to be able to save your VBA codes. If you do not save as xlsx, I will not be able to see your VBA code and you will lose points on those problems.
- Make sure you keep a copy of your work on your computer.
- Highlight your answers and make sure to label constants, initial values and equations. I need to follow your logic.
- Save each problem as different sheet in the same excel file. Screenshot your VBA code and paste it into excel.
- Good luck!

Question 1 (70 points, each subsection is 10 points):

In thermodynamics, an equation of state (EOS) is a thermodynamic equation relating state variables which describe the state of matter under a given set of physical conditions, such as pressure, volume and temperature. Equation of states are useful in describing properties of fluids. Currently there is no single EOS accurately predicts the properties of all substances under all conditions. Thus, different equations of states are derived and used in certain applications. Some example of EOS: ideal gas, Van der Waals and Peng-Robinson. The equation shown below is also one of the EOS.

$$p(V - b) = RTe^{-\frac{a}{RTV}}$$

a and b values can be calculated from T_c (critical temperature) and P_c (critical pressure).
 $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$

$$a = \frac{4R^2T_c^2}{e^2P_c} \quad b = \frac{RT_c}{e^2P_c}$$

For Ammonia $T_c = 406\text{K}$ and $P_c = 112.8 \text{ atm}$. Find the molar volume of ammonia at 300K and 7 atm by using;

- Plot molar volume vs
- Goal Seek
- Newton's method
- Secant Method
- Create a VBA function that will calculate the molar volume by using the EOS above

- f. Create a VBA Sub that will ask user to input temperature and pressure and calculate the molar volume by using EOS above and assigning that value into cell "E1"
- g. Calculate molar volume of ammonia at 7 bar between 20-200°C with increments of 20. Use automated Goal Seek macro.

Question 2 (30 points):

Xylene, styrene, benzene and toluene are to be separated with an array of distillation column as shown below. Calculate molar flow rates of D_1 , D_2 , B_1 , B_2 by using built-in matrix functions in Excel.

